

Response to periodic variations in precipitation rate

This derivation is based on the main assumption that perturbation in precipitation rate only induce perturbation in slope and do not induce perturbations in drainage area (no migration of the divide, for example).

Consider the linear form of the SPL, i.e., $n = 1$:

$$\frac{\partial h}{\partial t} = U - K A^m \nu_0^m S \quad (1)$$

where ν_0 is the precipitation rate, assumed to be uniform. By averaging across the mountain (i.e., along the spatial coordinate), we obtain an evolution equation for the mean slope of the mountain:

$$\frac{\partial \bar{h}}{\partial t} = L \frac{\partial \bar{S}}{\partial t} = U - K' \nu_0^m \bar{S} \quad (2)$$

where the area dependence has been moved into K' . L is the half-width of the mountain. When the system reaches steady-state, we can write:

$$0 = U - K' \nu_0^m \bar{S}_0 \quad (3)$$

where $\bar{S}_0$ is the mean steady-state slope of the mountain.

Now, consider a temporal perturbation in precipitation rate:

$$\nu = \nu_0 + \Delta\nu \quad (4)$$

which will cause a perturbation in slope around the steady-state value:

$$\bar{S} = \bar{S}_0 + \Delta\bar{S} \quad (5)$$

The averaged SPL (Equation 2) becomes:

$$L \frac{\partial(\bar{S}_0 + \Delta\bar{S})}{\partial t} = U - K'(\nu_0 + \Delta\nu)^m(\bar{S}_0 + \Delta\bar{S}) \approx U - K' \nu_0^m (1 + m \Delta\nu')(\bar{S}_0 + \Delta\bar{S}) \quad (6)$$

where, $\Delta\nu' = \Delta\nu/\nu_0$. Neglecting the second-order term, we get:

$$L \frac{\partial \Delta\bar{S}}{\partial t} \approx U - K' \nu_0^m \bar{S}_0 - K' \nu_0^m \Delta\bar{S} - K' \nu_0^m m \Delta\nu' \bar{S}_0 \quad (7)$$

which simplifies to:

$$L \frac{\partial \Delta\bar{S}}{\partial t} = -K' \nu_0^m \Delta\bar{S} - K' \nu_0^m m \Delta\nu' \bar{S}_0 \quad (8)$$

following equation 3. Scaling the equation by the steady-state mean slope, $\bar{S}_0$, to introduce the dimensionless variation in slope, $\Delta\bar{S}' = \Delta\bar{S}/\bar{S}_0$ leads to:

$$L \frac{\partial \Delta\bar{S}'}{\partial t} = -K'\nu_0^m \Delta\bar{S}' - K'\nu_0^m m \Delta\nu' \quad (9)$$

Finally, introducing the response time scale, $\tau = h_0/U$ to define a dimensionless time, i.e., $t' = t/\tau$, and considering that the steady state height, h_0 , is given by $h_0 = \bar{S}_0/L$, we get:

$$\frac{L}{K'\nu_0^m \tau} \frac{\partial \Delta\bar{S}'}{\partial t'} = \frac{\partial \Delta\bar{S}'}{\partial t'} = -\Delta\bar{S}' - m \Delta\nu' \quad (10)$$

Now, considering a dimensionless periodic variation in precipitation of the form:

$$\Delta\nu' = \Delta\nu'_0 \sin(\beta t') \quad (11)$$

where $\beta = 2\pi\tau/P$ and P is the period of the variation, Equation 10 becomes:

$$\frac{\partial \Delta\bar{S}'}{\partial t'} = -\Delta\bar{S}' - m \Delta\nu'_0 \sin(\beta t') \quad (12)$$

which has the following solution

$$\Delta\bar{S}' = -\frac{m\nu'_0}{\sqrt{1+\beta^2}} \sin(\beta t' - \pi/2 + \psi) \quad (13)$$

where $\psi = \arctan(1/\beta)$.

The flux of material coming out of the mountain of size L , is given by:

$$\phi = \int_L (U - \frac{\partial h}{\partial t}) dL = LU - L \frac{\partial \bar{h}}{\partial t} \quad (14)$$

This corresponds to the sum of a steady-state value, LU , and a deviation from steady-state, $\Delta\phi = -L \partial\bar{h}/\partial t$, which in a dimensionless form can be expressed as:

$$\Delta\phi' = \Delta\phi/LU = -\frac{1}{U} \frac{\partial \Delta\bar{h}}{\partial t} = -\frac{L}{U} \frac{\partial \Delta\bar{S}}{\partial t} = -\frac{Lh_0}{UL} \frac{\partial \Delta\bar{S}'}{\partial t} \quad (15)$$

and using the dimensionless time, t' , yields:

$$\Delta\phi' = -\frac{\partial \Delta\bar{S}'}{\partial t'} \quad (16)$$

from which we can derive the following expression for the sediment flux response to a periodic rainfall forcing:

$$\Delta\phi' = \frac{\beta m \Delta\nu'_0}{\sqrt{1+\beta^2}} \sin(\beta t' + \psi) \quad (17)$$